Planning of Obstacle-Aided Navigation for Multi-Legged Robots Using a Sampling-Based Method Over Directed Graphs

Kaustav Chakraborty, Haodi Hu, Matthew D. Kvalheim, and Feifei Qian, *Member, IEEE*

***Abstract*—Existing work in legged robot navigation in cluttered environments often seeks collision-free paths that avoid obstacle interactions. Here we present a new approach for multi-legged robots to utilize leg-obstacle collisions to generate desired dynamics. To predict the change of robot state under repeated leg-obstacle collisions, we construct a discretized directed graph model: each node of the graph represents a different robot state, whereas the directed edges pointing from one node to another represent the transitions from one robot state to the next within one stride. These obstacle-modulated state transitions can depend on the robot gaits used. To capture this dependence, an empirical interaction model is used to compute the change of robot state based on initial contact positions between robot legs and obstacles. To validate the prediction of robot state transitions, we experimentally measure the state of a quadrupedal robot as it traverses a periodic obstacle field with three different gaits: bound, trot, and pace. We observed that the robot could passively converge to different steady state orientations, and these steady states corresponded well with the Strongly-Connected-Path-Components (SCPCs) within the directed graph. Searching over the graph for connected paths of SCPCs allows development of a gait planner that can generate gait switching strategies for a robot to achieve desired states by simply engaging with a sequence of leg-obstacle collisions. We demonstrate in experiments that using this method an open-loop quadrupedal robot was able to achieve desired orientations within a periodic obstacle field without any sensory input or active steering.**

***Index Terms*—Dynamics, legged robots.**

I. Introduction

LEGGED robots are expected to traverse complex terrains — such as earthquake rubble, collapsed buildings with fallen columns, and environments with densely distributed large rocks and boulders (Fig. 1) — while accomplishing tasks like search and rescue, exploration, and construction [1], [2]. These terrains pose significant challenges to robot locomotion and navigation, as the large perturbations from leg-obstacle collisions could cause the robots to slip, get stuck, or even flip over. In most existing path planning techniques [3], [4], occupancy is used interchangeably with "obstacles," and the robot is required to find a "obstacle-free" path to navigate through the environment. This "obstacle avoidence" concept works well for wheeled robots or vehicles driving on flat terrains. However, for legged robots that are capable of traversing rocks and boulders with sizes smaller or comparable with their legs, a new representation of "obstacles" for navigating obstacle-cluttered terrains is required.

Fig. 1. Natural environments containing large obstacles such as rocks and boulders are challenging for legged locomotors. Effective locomotion and navigation on such terrains require robots (a) and animals (b) to assess outcomes of leg-obstacle interactions, and adjust locomotion strategies accordingly. Photos credit (a) ©Peter S (cc-by-sa/2.0) (b) cc-0 PublicDomain.

Recent advances in robotics [5]–[7] and biomechanics [8]–[11] suggested that animals and robots could effectively move through complex 3-dimensional environments by exploiting interactions with "obstacles". Cockroaches were observed to passively select low-effort pathways through cluttered vertical beams by adjusting their body yaw and pitch angles during obstacle collision [8]. Snakes and snake-like robots could push against vertical pegs to propel on slippery sand [7], [10]. Bio-inspired legged robots with different periodic gaits could robustly yield different headings under repeated obstacle collisions [5], [6]. These discoveries suggested the potential for legged robots to actively engage with previously-considered "obstacles" and take advantage of the large perturbation forces to aid with traversal of challenging terrains.

However, the question of how robots can select obstacle-engaging actions to generate desired dynamics remains challenging. This is partly due to the complexity in modelling the evolving robot dynamics under robot-obstacle collisions. These

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collisions can lead to large interaction forces during a short amount of time, resulting in a significant amount of uncertainty in control and state estimation. In addition, unlike flat ground with different heights [12] or discrete footholds [12], [13], robots running over a boulder field (Fig. 1(a)) could experience constant slipping or other relative movements between legs and obstacles during each stance, making it difficult to directly apply existing control and optimization methods that require static foot-ground contact [14]. As an attempt to address this challenge, in this study we integrate a physics-based robot-obstacle interaction model with a sampling based planning method, to allow legged robots to plan gait switching sequences that utilize obstacle interactions to generate desired dynamics.

The major contributions of this work are summarized as follows:

- Leveraging an empirical interaction model [6] to compute the change of robot state within one stride of each leg-obstacle collision, for three different gaits: bound, trot, pace.
- Using a discretized directed graph to represent robot state transitions under repeated leg-obstacle collision as it traverses a periodic obstacle field with the three gaits
- Using graph algorithmic techniques to identify structures called Strongly Connected Path Components (SCPCs) [15], and demonstrating that these SCPCs correspond well with leg-obstacle collision induced robot steady states characterized from experiments.
- Developing a gait sequence planner that can find gait switching sequences for multi-legged robots to achieve certain desired state sequences in obstacle fields simply by sequentially connecting SCPC states, and without any sensory input or body-level steering.

II. A Directed Graph Representation of Robot State Transitions in an Obstacle Field

In this section, we discuss how we model the state (i.e., position and orientation) change of a quadrupedal robot as it traverses a periodic obstacle field with three different gaits. Section II-A introduces the robot and the experiment setup. Section II-B describes the discretized directed graph representation of robot state transitions, and the formulation of the model that generates the graph. We show that by searching for SCPCs within the graph (Section II-C), we can identify steady-state orientations that an open-loop robot could passively converge to in the obstacle field simply by maintaining a periodic gait (Section II-D).

A. Physical System: Robot, Environment, and Gaits

The robot used in this study is a small RHex-class [16] quadrupedal robot (Fig. 2(a)). The robot body measured 16.5 cm long, 15.2 cm wide, 8.9 cm tall, and weighted 1.4 kg. The robot leg (PLA plastic) is a bio-inspired [16] C shape with a diameter of 6 cm.

We study the robot's horizontal-plane dynamics as it traverses a simplified, parameterized obstacle field: an array of evenly-spaced half-cylindrical obstacles (PVC plastic). To emulate challenging terrains where obstacles are large (*i.e.,* comparable with robot leg size) and fairly densely distributed, the diameter

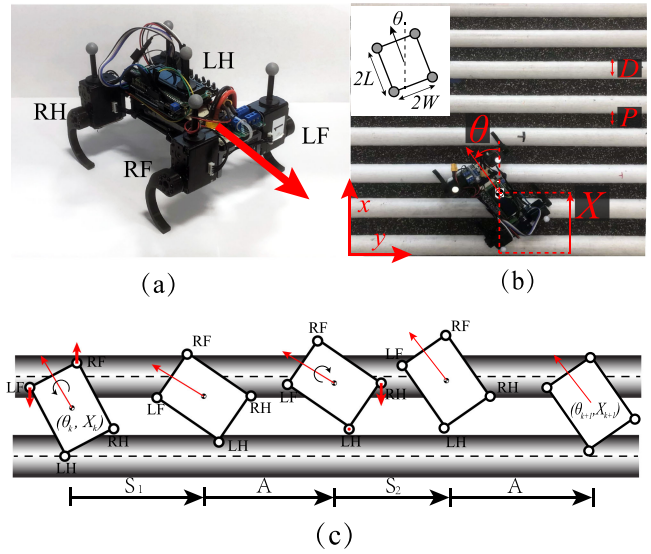

Fig. 2. Quadrupedal robot (a) and environment setup (b) used in our locomotion experiments, and the four discrete events (c) described in our model. In (a) and (c), LF, RF, LH, RH represent the four legs of the robot. (b) shows a top view of our simplified experiment setup: a quadrupedal robot traverses an array of half-cylindrical obstacles. X is the x-direction position of the robot Center of Mass (CoM). θ is the robot orientation. D is the obstacle diameter. P is the obstacle spacing. Inset: horizontal-plane representation of the robot in the "mode map" model. (c) depicts the four key events during each stride of leg-obstacle interaction for a bound gait. The four arrows, labeled S_1, A, S_2, A, represent the four discrete "modes": leg pair 1 dominated slipping, leg pair 1 dominated advancing, leg pair 2 dominated slipping, leg pair 2 dominated advancing, respectively.

of each obstacle, D, and the distance between obstacles, P, are chosen to be 4.8 cm and 4.0 cm, respectively (Fig. 2(b)). Due to the constant shape profile along the y axis, we describe the pose of the robot in the horizontal-plane (i.e., x-y plane) of the world frame as (θ, X), where θ is the robot's yaw angle measured counterclockwise from the $+x$-axis, and X is its Center-of-Mass (CoM) position along the x direction (Fig. 2(b)).

In an effort to understand how different locomotion strategies allow robots to exploit obstacle collisions and produce different dynamics, we study 3 different gaits of the quadrupedal robot: bound, trot and pace (Fig. 4 bottom diagram). In a bound gait, the left-front (LF) and right-front (RF) legs (Fig. 4 diagram, Bound, yellow legs) move synchronously, and alternate with the two hind legs, left-hind (LH) and right-hind (RH) (Fig. 4 diagram, Bound, green legs), with a half-cycle delay. In a trot gait, two diagonal legs, LF and RH (Fig. 4 diagram, Trot, yellow legs), move synchronously and alternate with the other pair, LH and RF (Fig. 4 diagram, Trot, green legs). In a pace gait, two right-side legs, RF and RH (Fig. 4 diagram, Pace, yellow legs), form one synchronous pair and alternate with the synchronous pair comprising the two left-side legs, LF and LH (Fig. 4 diagram, Pace, green legs). A fixed gait frequency of 0.33 Hz is used in this study.

B. Modelling Robot State Transitions During Obstacle Field Traversal

The goal of our model is to compute the updated robot state after one stride of each leg-obstacle collision, based on the robot state at the time of the collision, and the gait used within that

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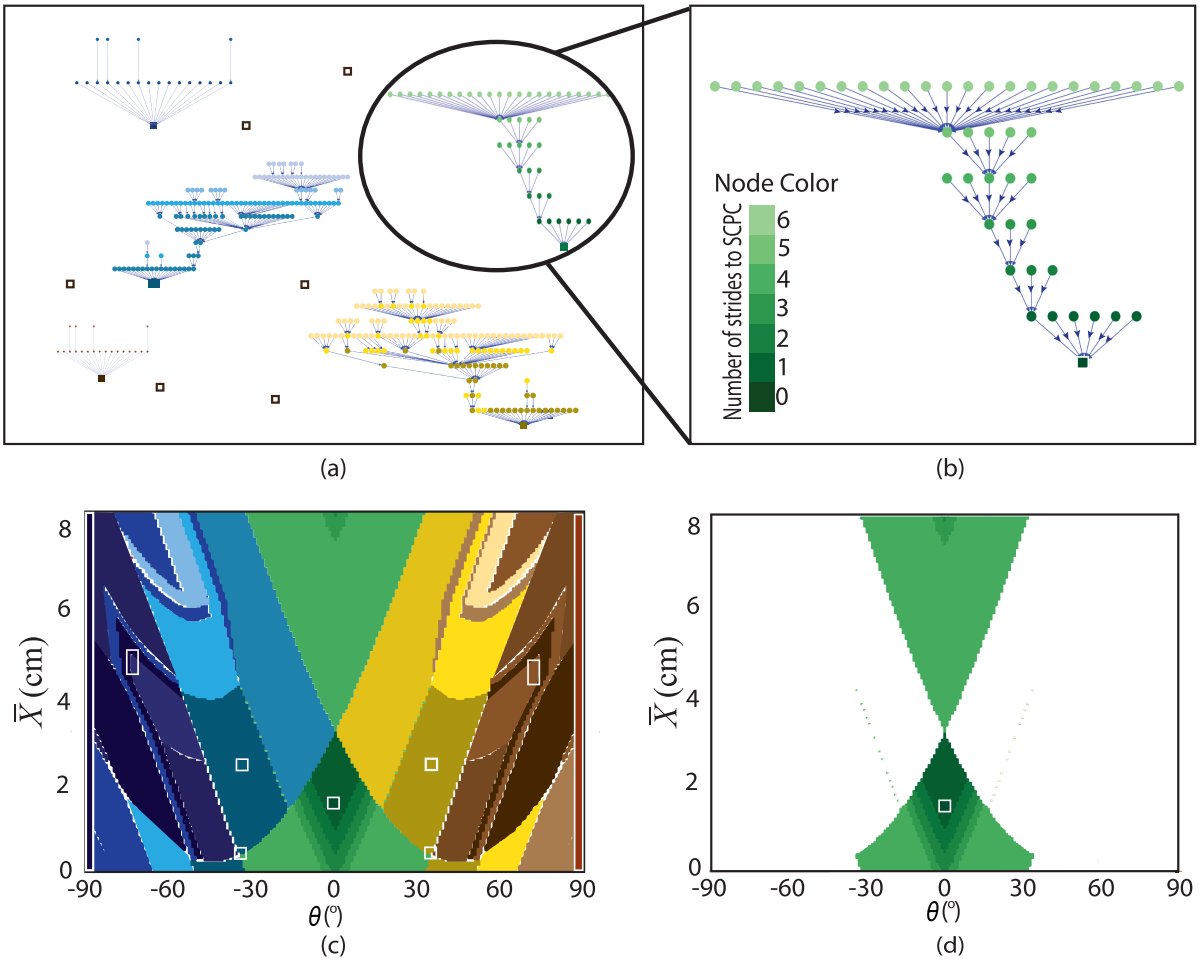

Fig. 3. Model-computed robot state transitions under repeated leg-obstacle collisions using a bound gait, visualized in directed graph representation (a, b) and in robot state space (c, d). (a) and (c) visualize the robot state space (θ, $\bar{X}$) for all discretized robot states with $\theta \in [-90°, 90°]$, whereas (b) and (d) visualize the discretized robot states within the basin of attraction of one specific SCPC node, (θ, $\bar{X}$) = (0°, 1.76 cm). In (a, b), nodes are represented with circles or squares (indicating SCPC nodes), and edges are represented with arrows. In (a, b, c, d), same color represents basins of attractions associated with the same SCPC node, or SCPC nodes with the same orientation. Shades of color of a node represent the number of transitions (*i.e.,* edges) between that node and SCPC nodes. Lighter color indicates a larger number of transitions for a node to converge to the SCPC.

stride.[1] To model the effect of leg-obstacle collisions on robot state change, we leverage an "obstacle disturbance selection" framework from [5]. Based on this framework, obstacles can be represented as "slipping disturbances" distributed in the world frame, and robot legs as disturbance "selectors" that select obstacle disturbances at their positions during the contact [5]. [5] and [6] demonstrated that given robot position and orientation upon leg-obstacle collision, the slipping disturbances at each leg's position can be computed using empirical measurements from [17]. Based on the slipping disturbances from each leg, updated robot position and orientation at the end of the step can be computed. Although the model in [6] was only applied to a trot gait, it was demonstrated to successfully capture robot steady-state orientations observed in experiments, and was useful in explaining how leg-obstacle contact positions affected robot orientations and trajectories. Since obstacle disturbances depend primarily on the inclination at leg-obstacle contact positions [17], a wide variety of obstacle shapes and distributions could be represented as a superposed obstacle disturbance field [5], enabling the analysis and planning methods based on the obstacle-disturbance-selection framework [5] to be extended to more complex environments.

Here we generalize this model to enable computation of robot state transitions for 3 gaits: bound, trot, pace. We approximate the robot body in the x-y plane as a rigid rectangle (Fig. 2(b) inset, Fig. 2(c)) with the same length and width as measured in experiments and a uniform mass distribution so that the CoM is at the geometric center of the rectangle. Four robot legs, LF, RF, LH, RH, are approximated as four vertices of the body rectangle (Fig. 2(c)). To model the three gaits, we use O_1 (Fig. 4 diagram, yellow legs) and O_2 (Fig. 4 diagram, green legs) to represent the two alternating leg pairs. Two legs within the same group move synchronously and alternate with the other pair. For bound gait, $O_1 = \{LF, RF\}$, $O_2 = \{LH, RH\}$; for trot $O_1 = \{LF, RH\}$

[1]One stride is the time cycle when a leg completes a full revolution. For the three gaits (bound, trot, pace) studied here, each stride contains two steps, *i.e.,* two touchdown events from the two synchronized leg pairs.

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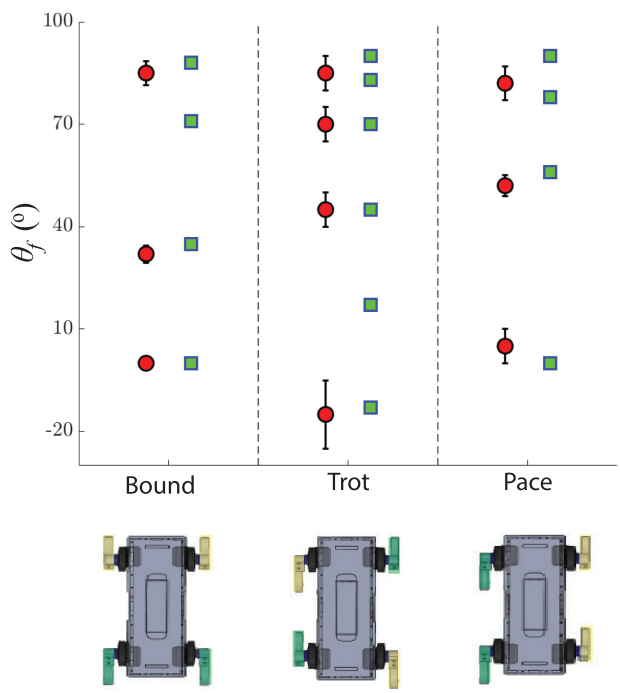

Fig. 4. Comparison of robot steady-state orientation between SCPCs identified from the directed graph (green squares) and experimental measurements (red circles), for bound, trot and pace gaits. Red circles represent averaged robot orientation over the last 5 seconds. Error bars represent the standard deviation of the averaged orientations. Bottom diagram illustrates the three different gaits. Yellow legs represents legs in synchronous pair O_1, and green represents legs in synchronous pair O_2.

and $O_2 = \{LH, RF\}$, and for pace $O_1 = \{RF, RH\}$ and $O_2 = \{LF, LH\}$.

For a given gait, we represent each stride cycle as a composition of four discrete events (which we refer to as "modes," Fig. 2(c) $S_1 \to A \to S_2 \to A$): (i) Leg pair 1 touches down and slips; (ii) Leg pair 1 advances the robot body forward during the remaining duration in stance; (iii) Leg pair 2 touches down and slips; (iv) leg pair 2 advances the robot body during the remaining of the stance. We define a gait-dependent return map [18], $F_0 : \mathbb{S}^1 \times \mathbb{R} \to \mathbb{S}^1 \times \mathbb{R}$, which sends the robot's pose at the k_{th} stride, (θ_k, X_k), to its pose at the $(k+1)_{th}$ stride, (θ_{k+1}, X_{k+1}). We define this return map, F_0, as a composition of four sub "mode maps" corresponding to the four discrete events: $F_0 := S_1 \circ A \circ S_2 \circ A$. The "slipping" mode maps, S_1 and S_2, correspond to discrete events (i) and (iii), compute the transition from robot pose prior to leg-obstacle contact at the k-th stride, (θ_k^-, X_k^-), to the robot pose right after obstacle interaction, (θ_k^+, X_k^+), based on the pre-slip leg-obstacle contact position [5], [6]. The "advancing" mode map, A, corresponding to discrete events (ii) and (iv), translates the robot CoM along its post-slipping orientation with a fixed step length, measured from experiments. Since the obstacle distribution in this study is periodic along the x direction, the mode maps S_1 and S_2 commute with $X \mapsto X + (P + D)$, where $P + D$ is the spatial period along the x direction (*i.e.,* the sum of obstacle spacing and obstacle diameter). Thus, F_0 descends to a well-defined map $F : \mathbb{T}^2 \to \mathbb{T}^2$, where $\mathbb{T}^2$ is the torus viewed as the rectangle $[-\pi, \pi] \times [0, P + D]$ with opposite edges identified. We therefore define the robot (reduced) state space as (θ, $\bar{X}$), where $\bar{X} := X \; mod \; (P + D)$ represents the relative robot X position within each spatial period.

To approximate the robot horizontal-plane dynamics during obstacle field traversal in a computationally useful way, we construct a directed graph to represent discrete robot state transitions from one stride to another upon repeated leg-obstacle collisions, given a certain gait. Each node of the directed graph (Fig. 3(a) and (b), circles) represents a robot state. Given a specific gait, each directed edge (Fig. 3(a) and (b), arrows) pointing from a parent node, n_p, to a child node, n_c, indicates that the robot would transit from state (θ_p, $\bar{X}_p$) to state (θ_c, $\bar{X}_c$) under the interaction forces with the physical environment.

To construct the directed graph, we perform the following steps:

1) discretize robot state space to obtain a grid[2];
2) create a node for each grid midpoint;
3) For each node v:
   a) for each gait, use the gait-dependent return map to compute $F(\theta, \bar{X})$, where $(\theta, \bar{X})$ is the state corresponding to v;
   b) for each gait, add an arrow from v to the node corresponding to one of the grid cells containing $F(\theta, \bar{X})$.

C. Directed-Graph Enabled Identification of Potential Steady States and Associated Basins of Attraction

The structure of the computed directed graph allows for a tractable algorithmic analysis of how variation in gaits could lead to different robot state transitions under the influences of obstacle collisions.

Fig. 3(a) visualizes the directed graph representing robot state transitions in the periodic obstacle field using a bound gait. We noticed that many of the upper-level nodes ended up converging to a single node at the bottom (Fig. 3(a), squares). The robot states corresponding to these "attracting" bottom nodes can correspond to equilibrium points, or to periodic orbits, or, a priori, to more complicated attractors.

The observed "convergence" of upper-level robot states towards these equilibria or periodic orbits in our directed graph was fairly consistent with observations from experiments in our study (Section II-D) and previous studies [5], [6]: with a periodic gait, a robot traversing over a periodic obstacle field could passively converge to different steady-state orientations under leg-obstacle collisions and without any body-level steering.

We hypothesize that by identifying these "attracting" nodes in our directed graphs, we could predict these experimentally-observed robot steady-state orientations, and use these predictions to guide the robot's obstacle engagement planning (Section III). Inspired by mathematical work on the computer-assisted study of dynamical systems [15], we identify candidate steady states by computing the Strongly Connected Path Components (SCPCs) [16, Def. 2.33] of our directed graph. SCPCs are related to (but distinct from) the standard graph-theoretic notion

[2]The graph used in this study is with a discretization of 180 × 180 grid, corresponding to a resolution of 1° in θ and 0.05 cm in $\bar{X}$. Simulation results suggested that the graph structure remains qualitatively similar with higher-resolution discretization.

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of Strongly Connected Components (SCCs) [16, Def. 2.31]: by definition, an SCC is a maximal set of vertices such that, for every pair of vertices A, B in the set, there is a directed path from A to B and vice versa; an SCPC is an SCC that contains at least one edge. Previous research has established that, when the directed graph is a carefully-constructed "outer approximation" [15] of the dynamical system, the union of grid cells in state space corresponding to an SCPC yield a "Morse decomposition" for the nondiscretized dynamics [16, Thm 2.25]. Examples of "Morse sets" generally correspond to asymptotically stable equilibria and periodic orbits, among other things. Therefore, we posit that the SCPCs of our directed graphs might roughly represent some Morse sets that correspond to equilibria or periodic orbits of our system. Therefore, here we search for SCPCs in our directed graph to identify potential robot steady states. We use Tarjan's algorithm [19] to find the strongly connected components (SCCs) in our graph. Then the SCPCs can be identified by removing all the singleton SCCs that do not have a self-mapping edge. In Section II-D we compare the identified SCPCs from our graph with experimentally-observed robot steady states.

The SCPCs in the directed graph not only predict potential robot steady states under leg-obstacle collisions, but also enable the prediction of potential robot states that will get "attracted" towards these steady states. Fig. 3(b) shows a visualization of a specific SCPC (dark green square) and all its parent nodes from different levels (lighter green circles). Since each arrow from a parent node to a child node represents robot state transition within one stride cycle, under the modulation of leg-obstacle collisions a robot starting from any of the lighter-color parent nodes would converge towards the SCPC state within limited number of strides. These parent nodes of an SCPC, therefore, forms the "basin of attraction" of the equilibrium state associated with the SCPC node.

Embedding the set of nodes of the directed graph (Fig. 3(a) and (b)) in the robot's state space (Fig. 3(c) and (d)) gives us a convenient procedure for understanding the evolution of the dynamics for each state. Each of the color schemes in Fig. 3(c) depicts a basin. If we start from any of the lighter shades of a particular colored basin and apply the dynamics, we will end up in a state having a darker shade of the same color. Within a finite number of stride cycles we will end up in the steady states of the basin having the darkest shade of the corresponding color (and specifically marked by white squares). This basin can be thought of as a squashed down 2D representation of the directed graph. We can actually think of each basin having the same levels as its corresponding tree, giving rise to the concept of height of the basin. This visualization of directed graph on the robot state space is useful in revealing the range of each basin, and the boundary between different basins, which is important in guiding a robot's planning of obstacle engagement. For example, the distance to basin boundary could be used to approximate the sensitivity to noise to maintain a steady state. If a SCPC state is located far from the basin boundary (e.g., Fig. 3(c), green square), even if the robot drifted to a nearby state under sensor or actuator noise, as long as it is still within the same attracting basin (e.g., Fig. 3(c), green region), the environment interaction forces will push the robot back towards the SCPC state (*i.e.,* the "lowest energy point" in the basin map). We demonstrate in Section III that, with the knowledge of gait-dependent SCPCs and associated basins, a robot can select specific gait sequences to ensure convergence to a desired SCPC state (Fig. 4). By sequentially visiting different SCPC states, the robot could potentially take advantage of the passive dynamics to robustly move along a desired trajectory.

D. Comparison of SCPCs Identified From the Directed Graph With Experimentally-Observed Robot Steady States

To compare the SCPC nodes identified from our directed graph with experimentally-observed robot steady states, we performed 144 experiments, 48 experiments for each robot gait, using the setup described in Section II-A. For each gait, robot initial orientation, θ_0, was systematically varied from 0° to 90° with an increment of 5°. Initial position of the robot in the world frame, X_0, was fixed at 0 cm for all trials. 3 trials were performed for each initial condition, (θ_0, X_0). For all trials, the robot simply maintains a periodic gait without any body-level steering. The robot position and orientation during each trial were tracked with a motion capture system (4x Optitrack Prime 13 W) with a frame rate of 120 FPS.

Our experiment demonstrates that, without any active steering, the disturbance forces from the leg-obstacle collisions can drive the robot to converge to a number of steady-state orientations (Fig. 4, red circles). The robot with a bound gait could passively stabilize at orientation angles 0°, $32^\circ \pm 2^\circ$ and $85^\circ \pm 3^\circ$; with a trot gait the robot could passively stabilize at orientation angles $-15^\circ \pm 10^\circ$, $45^\circ \pm 5^\circ$, $70^\circ \pm 5^\circ$ and $85^\circ \pm 5^\circ$; and a pace gait could passively stabilize at orientation angles $5^\circ \pm 5^\circ$, $52^\circ \pm 3^\circ$ and $82^\circ \pm 5^\circ$. These gait-dependent steady-state orientations provide the possibility for an open-loop robot to robustly navigate cluttered obstacle fields: by switching between different gaits, the leg-obstacle collisions will drive the robot towards different orientations.

Fig. 4 shows that most of the experimentally-observed steady states were well captured by the SCPCs identified from the directed graph (Fig. 4, green squares). We note that there are a few SCPCs that were not corresponding to experimentally-observed steady states. This is possibly due to a small basin of attraction associated with these SCPCs, in which case these steady states can be highly sensitive to noise, resulting in low probability for these steady states to be observed experimentally. Despite these small discrepancies, we demonstrate in Section III that the capabilities of the directed graph in capturing potential robot steady states and revealing the basins of attraction of these steady states can allow an open-loop robot to estimate and utilize leg-obstacle collision outcomes to achieve desired states within obstacle fields.

III. Directed Graph Based Gait Planning for Obstacle-Aided Navigation

The gait-dependent robot steady state orientations (Fig. 4) provides an opportunity for robots to exploit passive dynamics to

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effectively move towards desired locations: each gait would allow the robot to passively converge to different attracting steady states, and by switching between different gaits, a robot could sequentially visit a sequence of attracting states to move towards desired region. For robots with limited sensing and actuation resources, exploiting such passive dynamics to move through cluttered environments could potentially be more efficient and robust than actively cancelling out collisions dynamics, as the environmental interaction forces have a tendency to push the robot towards the attracting steady states. In this section, we use the predicted attracting steady states from the directed graph to develop a gait transition planner that could guide the autonomous selection of gait sequences for robots to exploit passive dynamics for navigation.

A. Gait Planner Implementation

Our planner integrates the obstacle-modulated robot state dynamics and the Rapidly exploring Random Trees (RRT) algorithm [20] to find robot gait sequences that could lead to desired state dynamics under leg-obstacle collisions. RRT is a single-query sampling based method that is commonly used in path planning [21] tasks. In traditional path-planning applications, the robot is treated as a point mass that can move towards any direction, and the goal of the algorithm is to find an obstacle-free path from a starting location to a goal location. In this study, we adopt the RRT to query the directed graph for gait sequences that could allow an open-loop controlled robot to achieve desired orientation state by "selecting" obstacle disturbances.

Given a task with the initial state of the robot, $(\bar{X}_0,\ \theta_0)$, and the desired goal orientation of the robot, θ_g, we use the RRT algorithm to search for gait transitions that connect the robot initial state to the desired goal state through the SCPCs in the directed graph (Algorithm 1). During the exploration phase of RRT, we use the the pre-computed directed graph (DirGraph) to explore the robot state space. Specifically, during each iteration the function NextState (Algorithm 1, line 5) uses the directed graph to identify all SCPC nodes that are child nodes to the current robot state using the three gaits. This allows the algorithm to determine all possible gait choices that the robot could use to passively converge from one steady state to the next. The identified SCPC nodes provide multiple posterior states that the robot could choose to converge to by using one of the three different gaits. We add all these posterior states to the SearchGraph with the corresponding gait as edges. Once the algorithm explores for a fixed number of iterations (Algorithm 1, maxIters), the function AllSequences (Algorithm 1, line 11) returns all gait sequences that could enable the robot to passively converge from the initial state to the desired goal state.

B. Physical Experiments of Obstacle-Aided Navigation Using Planner-Generated Gait Transitions

The physics model informed directed graph allows prediction of attracting states and their associated basins. We have showed that by executing a periodic gait, a robot could passively converge to specific orientations (Section II-D) corresponding to the gait-dependent attracting states. Here we show that by sequentially composing these attracting states, we could send a open-loop robot from specified initial orientation to desired goal orientation without any sensory feedback.

Algorithm 1: Planner.

```
Require: start := starting state
Require: goal := goal orientation
Require: maxIters := max iterations for the exploration
Require: DirGraph := directed graph from Section II
1:  Add node start to SearchGraph, iter ← 0    ▷ initialize
2:  while iter < maxIters do
3:    target ← RandomState
4:    nearest ← NearestState(SearchGraph,target)
5:    next ← NextState(DirGraph, nearest)
6:    for each state ∈ next do
7:      if state ∉ SearchGraph then
8:        add node state to SearchGraph
9:        add edge (nearest,state) to SearchGraph
10:   iter ← iter+1
11: GaitSequence = AllSequences(SearchGraph, start,
    goal)
12: return GaitSequence
```

To demonstrate this, we use the gait planner to generate gait sequences that allow the robot to transit from the initial orientation, $(\theta_0,\ \bar{X}_0) = (50^\circ,\ 0\ \text{cm})$, to two different goal orientations, $\theta_{g1} = 0^\circ$ (Task 1) and $\theta_{g2} = 90^\circ$ (Task 2), by exploiting obstacle collision forces only. We allow the planner to explore for 500 iterations which took approximately 11 seconds to compute, and then output all sequences that requires fewer than 3 gait switches to converge to a SCPC state with desired goal orientation. The planner was able to find three sequences of gait switches for Task 1: Bound $\rightarrow$ Pace (BP), Bound $\rightarrow$ Trot $\rightarrow$ Pace (BTP), and Pace $\rightarrow$ Bound $\rightarrow$ Pace (PBP); and two gait sequences for Task 2: Pace $\rightarrow$ Trot $\rightarrow$ Bound (PTB), and Trot $\rightarrow$ Bound (TB). We implemented all five gait sequences in experiments, with 3 trials for each gait sequence. No sensory feedback was provided to the robot, and the robot simply switches to the next gait in a planned sequence every 9 seconds.[3]

Fig. 5 illustrates one of the planned gait sequences for Task 1: Pace $\rightarrow$ Bound $\rightarrow$ Pace. The planner predicted that from the initial state, $\theta_0 = 50^\circ$ (Fig. 5(a) and (b), I), with a pace gait the robot would passively converge towards 56° (Fig. 5(a) and (b), II) under the influence of obstacle disturbances. Subsequently with a bound gait, the robot would passively converge to the 35° SCPC (Fig. 5(a) and (b), III), and another pace gait would eventually make the robot converge from 35° to the desired goal orientation for Task 1, $\theta_{g1} = 0^\circ$ (Fig. 5(a) and (b), IV). Experimental image sequences (Fig. 5(c), I–IV) and experimentally-tracked robot orientation trajectories Fig. 5(d) corroborated with the predicted robot state transitions by the gait planner, and demonstrated that the robot robustly converged to each planner-predicted SCPC state (either fixed points or periodic oscillations), simply

[3]We record the last gait for 12 seconds (4 gait cycles) to obtain an accurate measurement of averaged final orientation.

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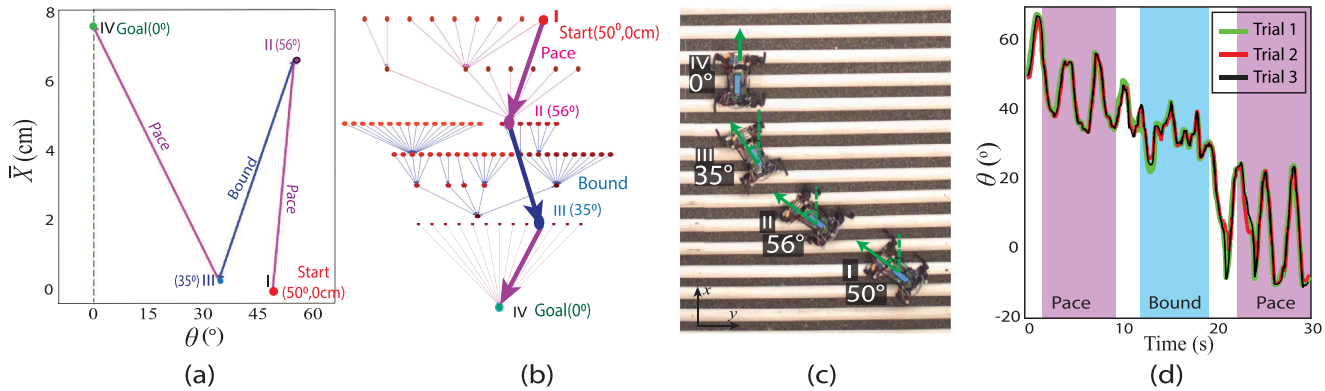

Fig. 5. One of the planner-generated robot gait sequence, Pace (purple) → Bound (blue) → Pace (purple), and predicted robot SCPC nodes connection, I (robot initial state, red) → II → III → IV (Task 1 goal, green), illustrated in robot configuration space (θ, $\bar{X}$) (a) and directed graph representation (b). Image sequences of robot key state corresponding to the predicted SCPC nodes (I, II, III, IV) from experiments are shown in (c). Robot orientation trajectories measured from the three trials for this gait sequence are shown in (d).

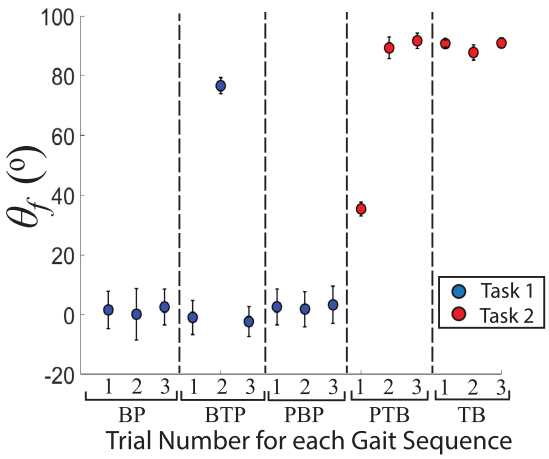

Fig. 6. Experimentally-measured robot final orientation angle for each gait sequence tested. 3 trials were performed for each gait sequence in Task 1 (blue) and Task 2 (red). Markers represent the converged orientation angle averaged over the last 6 seconds. Error bars represent the amount of periodic oscillation around the converged orientation angle.

by performing the planned gaits in sequence and without any sensory feedback or additional steering effort.

To evaluate the robustness of our method in enabling robots to exploit passive interaction dynamics for achieving desired orientations, we experimentally measured the averaged orientation angle of the robot during the last 6 seconds of the 15 trials. We show that the averaged orientation of the robot passively "locked" to the goal orientations, 0° and 90° under repeated leg-obstacle collisions, in 8 out of 9 trials for Task 1, and 5 out of 6 trials for Task 2 (Fig. 6). The reason behind the two trials that converged to different SCPC orientations is likely associated with the uncontrolled robot position at gait switching. Since the gait switches was executed at fixed time intervals, the robot state could be at a $\bar{X}$ position that belongs to the basin of another SCPC node. This issue could be addressed by using the basin of attraction map to guide the selection of exact gait-switching timing, and ensuring that the robot arrives at the desired basin. Furthermore, the distance to basin boundary could be used to assess and compare the robustness between potential gait sequences to further improve robustness. For example, a robot could use select SCPC states located far from the basin boundary to enhance insensitivity to state uncertainty. Future work should explore how to integrate this sensitivity assessment in the gait planner to enable balancing between robustness (e.g., larger SCPC basin) and efficiency (e.g., fewer gait transitions).

For the trials that converged to the predicted SCPC state, the final orientation consistently locked to the desired average steady-state angle: $0.5 \pm 1.0°$ for the 8 trials in Task 1, and $88.7 \pm 0.9°$ for the 5 trials in Task 2. In addition, orientation trajectories from the same gait sequence exhibited high consistency between different trials (Fig. 5(d)). The striking consistency, especially considering the robot does not have access to any sensing feedback and is constantly subject to repeated leg-obstacle collisions, demonstrated the robustness of our method, and opened up a new possibility: with a better understanding of the robot-environment interaction physics, even simple, low-cost robots could potentially exploit interactions with their physical environment to effectively move through the complex world.

IV. Conclusion and Future Directions

Our study offers a new method for legged robots to select gait sequences that can allow utilization of leg-obstacle collisions to achieve desired states. Based on leg-obstacle interaction physics observed from locomotion experiments, we developed a discrete-time return map that models robot state transitions from one stride to the next upon each leg-obstacle collision, for three different gaits: bound, trot, and pace. Using the gait-dependent return map, a directed graph representation was constructed to identify "attracting" SCPC nodes under repeated leg-obstacle collisions. The SCPC nodes identified from the graph agreed well with experimentally-characterized robot steady states, and enabled a sampling-based planning algorithm to generate gait switching sequences that can allow an open-loop robot to achieve desired state by utilizing leg-obstacle collisions and without any sensory feedback.

Our study provides a new perspective on utility of obstacle collisions, and is a key step in establishing obstacle-aided

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locomotion capabilities for resource-constrained robots to effectively move through challenging and cluttered environments. Although here we only demonstrated the use of three specific gaits to utilize obstacle disturbances, the planner could be generally applied to search for other locomotion strategies for obstacle-aided locomotion and navigation, such as robot step length, leg touchdown position [6], and additional gaits available to multi-legged robots. Another future direction is to extend our method to non-periodic environments. The structured environment used in this study allowed us to systematically test and compare the effect of different gaits on robot-obstacle interaction dynamics, and derive reduced-order model of robot state transitions. For future work to extend the method to more complex scenarios, non-periodic obstacle fields could be represented as a superposed disturbance force field [5] to allow predictions of robot dynamics for a wider variety of obstacle shapes and distributions. As a starting point of these promising future avenues, we believe that the robot-obstacle interaction dynamics modelling and gait planning methods proposed here could illuminate many possibilities for robots to take advantage of physical interactions with their environments for effective locomotion and navigation.

References

[1] J. Casper and R. Murphy, "Human-robot interactions during the robot-assisted urban search and rescue response at the world trade center," *IEEE Trans. Syst., Man, Cybern., Part B. (Cybern.)*, vol. 33, no. 3, pp. 367–385, Jun. 2003.

[2] A. Stroupe, T. Huntsberger, A. Okon, H. Aghazarian, and M. Robinson, "Behavior-based multi-robot collaboration for autonomous construction tasks," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2005, pp. 1495–1500.

[3] A. Elfes, "Using occupancy grids for mobile robot perception and navigation," *Computer*, vol. 22, no. 6, pp. 46–57, Jun. 1989.

[4] H. P. Moravec, "Sensor fusion in certainty grids for mobile robots," in *Sensor Devices and Systems for Robotics*, Berlin, Germany: Springer, 1989, pp. 253–276.

[5] F. Qian and D. E. Koditschek, "An obstacle disturbance selection framework: Emergent robot steady states under repeated collisions," *Int. J. Robot. Res.*, vol. 39, no. 13, pp. 1549–1566, 2020.

[6] D. Ramesh, A. Kathail, D. E. Koditschek, and F. Qian, "Modulation of robot orientation via leg-obstacle contact positions," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 2054–2061, Apr. 2020.

[7] J. M. Rieser *et al.*, "Dynamics of scattering in undulatory active collisions," *Phys. Rev. E*, vol. 99, no. 2, 2019, Art. no. 022606.

[8] C. Li, A. O. Pullin, D. W. Haldane, H. K. Lam, R. S. Fearing, and R. J. Full, "Terradynamically streamlined shapes in animals and robots enhance traversability through densely cluttered terrain," *Bioinspiration Biomimetics*, vol. 10, no. 4, 2015, Art. no. 046003.

[9] R. Othayoth, G. Thoms, and C. Li, "An energy landscape approach to locomotor transitions in complex 3D terrain," *Proc. Nat. Acad. Sci.*, vol. 117, no. 26, pp. 14987–14995, 2020.

[10] P. E. Schiebel, J. M. Rieser, A. M. Hubbard, L. Chen, and D. I. Goldman, "Collisional diffraction emerges from simple control of limbless locomotion," in *Proc. Conf. Biomimetic Biohybrid Syst.*, 2017, pp. 611–618.

[11] A. A. Transeth, R. I. Leine, C. Glocker, K. Y. Pettersen, and P. Liljebäck, "Snake robot obstacle-aided locomotion: Modeling, simulations, and experiments," *IEEE Trans. Robot.*, vol. 24, no. 1, pp. 88–104, Feb. 2008.

[12] M. Neunert, F. Farshidian, A. W. Winkler, and J. Buchli, "Trajectory optimization through contacts and automatic gait discovery for quadrupeds," *IEEE Robot. Automat. Lett.*, vol. 2, no. 3, pp. 1502–1509, Jul. 2017.

[13] M. Posa, C. Cantu, and R. Tedrake, "A direct method for trajectory optimization of rigid bodies through contact," *Int. J. Robot. Res.*, vol. 33, no. 1, pp. 69–81, 2014.

[14] I. Mordatch, E. Todorov, and Z. Popović, "Discovery of complex behaviors through contact-invariant optimization," *ACM Trans. Graph.*, vol. 31, no. 4, pp. 1–8, 2012.

[15] K. Mischaikow, "Topological techniques for efficient rigorous computation in dynamics," *Acta Numerica*, vol. 11, pp. 435–477, 2002.

[16] U. Saranli, M. Buehler, and D. E. Koditschek, "RHEX: A simple and highly mobile hexapod robot," *Int. J. Robot. Res.*, vol. 20, no. 7, pp. 616–631, 2001.

[17] F. Qian and D. I. Goldman, "The dynamics of legged locomotion in heterogeneous terrain: Universality in scattering and sensitivity to initial conditions," in *Proc. Robot.: Sci. Syst.*, 2015, pp. 1–9.

[18] M. Brin and G. Stuck, *Introduction to Dynamical Systems*. Cambridge, U.K.: Cambridge Univ. Press, 2002.

[19] R. Tarjan, "Depth-first search and linear graph algorithms," *SIAM J. Comput.*, vol. 1, no. 2, pp. 146–160, 1972.

[20] S. M. LaValle *et al.*, "Rapidly-exploring random trees: A new tool for path planning," *Annu. Res. Rep.*, pp. 98–11, 1998.

[21] M. M. Costa and M. F. Silva, "A survey on path planning algorithms for mobile robots," in *Proc. IEEE Int. Conf. Auton. Robot Syst. Competitions*, ICARSC, 2019, pp. 1–7.

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